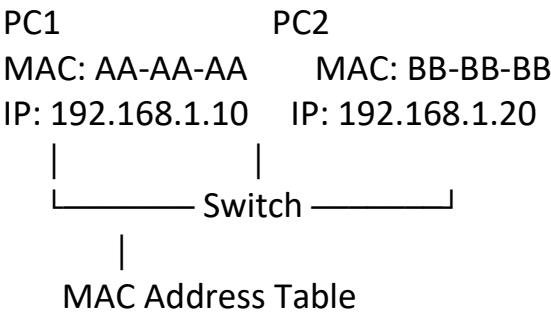


Question: Explain how the switch populates its MAC address table and forwards frames between the two devices, including what happens before the table is populated.

When **two devices are connected to the same switch** and communicate for the first time, the switch uses **MAC addresses** to learn where each device is located.

Example topology



1. Initially, the MAC table is empty

When the switch starts, it doesn't know which MAC address belongs to which port.

MAC Address Table

MAC Address	Port

(empty)	

2. PC1 sends a frame

Suppose PC1 wants to communicate with PC2.

PC1 sends an Ethernet frame:

Source MAC = AA-AA-AA
Destination MAC = BB-BB-BB

The frame enters the switch through **Port 1**.

3. Switch learns PC1's MAC

The switch looks at the **source MAC address**.

It learns:

"AA-AA-AA came from Port 1."

So it adds:

MAC Address	Port

AA-AA-AA	Fa0/1

This process is called **MAC address learning**.

4. But the destination MAC is unknown

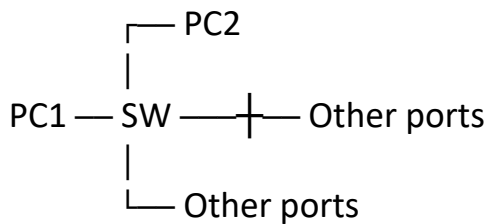
The switch checks:

Destination = BB-BB-BB

But BB-BB-BB isn't in its MAC table yet.

So the switch doesn't know which port PC2 is connected to.

It performs **unknown unicast flooding**.



The switch sends the frame out of the other appropriate ports **except the port where the frame arrived**.

PC2 receives the frame and processes it if the destination MAC matches.

5. PC2 replies

PC2 sends a reply back to PC1:

Source MAC = BB-BB-BB
Destination MAC = AA-AA-AA

The switch receives this on PC2's port and learns:

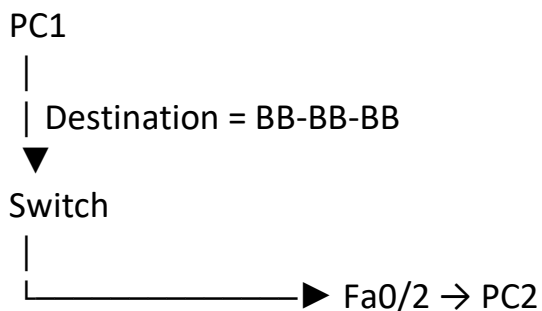
MAC Address	Port

AA-AA-AA	Fa0/1
BB-BB-BB	Fa0/2

Now the switch knows where both devices are.

6. Future communication is unicast

If PC1 sends another frame to PC2:



The switch looks up BB-BB-BB in its MAC table and sends the frame **only through Fa0/2**.

It does **not flood the frame** to every port.

A switch learns MAC addresses by examining the source MAC address of incoming Ethernet frames and associating that MAC address with the incoming port. When a destination MAC is not yet in the MAC address table, the switch floods the frame out of all ports except the incoming port. Once the destination device responds, the switch learns its MAC address as well. After both MAC addresses are learned, the switch forwards subsequent frames directly to the correct port instead of flooding them.